Matrix-Free Second-Order Orbital Optimization for Valence Bond Self-Consistent Field Theory

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ABSTRACT

Orbital optimization in valence-bond self-consistent-field theory must accommodate nonorthogonal orbitals, prescribed atomic-orbital supports, and relaxation of the structure coefficients. We formulate a matrix-free norm-extended optimization (NEO) treatment in support-preserving nonredundant coordinates, retaining orbital–structure coupling through independently expanded orbital and structure response spaces. The numerical study will compare NEO with conventional and block-preconditioned L-BFGS, with truncated Newton as an additional second-order comparison, reporting convergence, operator work, wall time, and peak memory under matched conditions. [Replace this sentence with the principal quantitative findings and the supported conclusion after the benchmark campaign.]

1 INTRODUCTION

1.1 The optimization problem

Explain why VBSCF uses nonorthogonal localized orbitals and, in HAO calculations, fixed orbital supports. Separate the cost of optimizing orbitals from the cost of solving the structure-coefficient problem. Introduce OEO as full-AO support, not as an orthogonal-orbital model.

1.2 Relation to second-order MCSCF optimization

Review restricted-step Newton, augmented-Hessian and norm-extended optimization, and matrix-free CASSCF implementations [1–3]. Identify what carries over to VBSCF—the local coupled quadratic model and iterative operator actions—and what does not carry over without derivation: orthogonal orbital rotations and their redundancy rules.

1.3 Scope of this work

State the methodological contribution as a second-order optimization framework for the specified nonorthogonal VB variational space. Treat integral representation and string contraction as supporting implementations, not independent novelty claims in this paper. The principal numerical question is when additional curvature information improves convergence enough to repay its cost relative to block-LBFGS.

2 THEORY

2.1 Nonorthogonal VB wave functions and the stationary structure problem

A VB wave function is expanded in a prescribed set of spin-adapted structures,

$$|\Psi_s(\mathbf{x})\rangle = \sum_{I=1}^{N_{\mathrm{str}}} a_{Is}|\Phi_I(\mathbf{x})\rangle. \tag{1}$$

The packed vector $\mathbf{x}$ contains the allowed AO coefficients of the inactive and active orbitals. The structures need not be orthogonal. We denote their overlap and Hamiltonian matrices by $\mathbf{S}$ and $\mathbf{H}$, respectively, and reserve $\mathbf{S}_{\mathrm{AO}}$ for the AO overlap. Here $\mathbf{H}$ includes the inactive-electron contribution; an equivalent implementation may separate a common reference energy from the active Hamiltonian. The nuclear repulsion is independent of $\mathbf{x}$ and can be added afterward.

At fixed orbitals, variation of the structure coefficients gives

$$\mathbf{H}\mathbf{a}_s = E_s\mathbf{S}\mathbf{a}_s, \qquad \mathbf{a}_r^T\mathbf{S}\mathbf{a}_s = \delta_{rs}, \qquad \mathcal{E}(\mathbf{x}) = \frac{1}{m}\sum_{s=1}^{m} E_s(\mathbf{x}). \tag{2}$$

We consider real wave functions and equal weights for the lowest m states; $m = 1$ gives ground-state VBSCF. The following derivation assumes a positive-definite structure overlap on the working coefficient space and a selected eigenspace separated from excluded roots. Internal degeneracies among the equally weighted states do not change the state-averaged energy. A gap closing between selected and excluded roots, however, can destroy differentiability of this objective.

For an orbital direction, let a dot denote the derivative at the current accepted point. Stationarity with respect to the structure coefficients gives

$$\dot{\mathcal{E}} = \frac{1}{m}\sum_{s=1}^{m} \mathbf{a}_s^T(\dot{\mathbf{H}} - E_s\dot{\mathbf{S}})\mathbf{a}_s. \tag{3}$$

Thus no coefficient-response solve is required for the first derivative. Coefficient response is required for the second derivative: the cancellation leading to eq (3) does not permit the eigenvectors to be held fixed when that equation is differentiated.

2.2 Support-preserving orbital coordinates and the update map

For orbital i, an injection matrix $\mathbf{R}_i$ embeds its allowed coefficients into the AO space, $\mathbf{c}_i = \mathbf{R}_i\mathbf{x}_i$. A localized HAO orbital has a prescribed subset of allowed AOs; an OEO orbital has full AO support. Neither designation imposes mutual orthogonality of the VB orbitals.

We remove a direction only when it is an infinitesimal invariance of the represented wave function and respects the specified supports. These invariances include orbital scaling and admissible changes of inactive representatives. For a complete active-space expansion with full-AO orbitals, nonsingular changes of the active basis can also be compensated by changes of the structure coefficients. This active-space invariance does not generally hold for an arbitrary selected set of VB structures. Consequently, active–active mixing is not removed merely because both orbitals are active.

The support condition must be imposed on the gauge generators themselves. If $\mathbf{G}_i$ collects candidate gauge generators for orbital i, and $\mathbf{F}_i$ extracts its forbidden AO rows, admissible gauge combinations satisfy

$$\mathbf{F}_i \mathbf{G}_i \mathbf{t} = \mathbf{0}. \tag{4}$$

The complement of these admissible generators defines a local quotient basis. In particular, truncating a dense gauge generator to the allowed rows is not equivalent to eq (4). Local metric conditioning of the quotient basis produces a lift $\mathbf{U}_k$ from reduced coordinates $\mathbf{p}$ to packed raw coefficients.

The orbital update and the preparation of orbitals for energy evaluation are distinct maps:

$$\mathbf{x}(\mathbf{p}) = \mathbf{x}_k + \mathbf{U}_k \mathbf{p}, \qquad \boldsymbol{\theta}(\mathbf{p}) = \mathcal{P}(\mathbf{x}_k + \mathbf{U}_k \mathbf{p}). \tag{5}$$

Here $\mathcal{P}$ includes orbital normalization and inactive-space projection used by the objective, together with admissible representative choices. The lift is held fixed within a local model and rebuilt after an accepted step. Equation (5) is an additive coefficient update followed by preparation, not an orthogonal orbital rotation or a sphere exponential map.

All derivatives must refer to this same composition. For a scalar function F of prepared orbitals, its Hessian pullback is

$$D^2(F \circ \boldsymbol{\theta})[\mathbf{u}, \mathbf{v}] = D^2 F[D\boldsymbol{\theta}\mathbf{u}, D\boldsymbol{\theta}\mathbf{v}] + DF[D^2\boldsymbol{\theta}[\mathbf{u}, \mathbf{v}]]. \tag{6}$$

The second term is necessary even though the raw lift in eq (5) is linear. For example, normalization of a raw orbital gives

$$\bar{\mathbf{c}} = \frac{\mathbf{c}}{n}, \quad n = (\mathbf{c}^T \mathbf{S}_{\mathrm{AO}} \mathbf{c})^{1/2}, \quad D\bar{\mathbf{c}}[\delta\mathbf{c}] = \frac{(\mathbf{I} - \bar{\mathbf{c}}\bar{\mathbf{c}}^T \mathbf{S}_{\mathrm{AO}})\delta\mathbf{c}}{n}. \tag{7}$$

Using an orbital-rotation Hessian with this coefficient update would omit the required map derivatives.

2.3 Physical orbital step metric

The trust-region norm measures changes of physical orbital representatives rather than the Euclidean magnitude of raw AO coefficients. Let $\mathcal{L}_k$ map a packed coefficient tangent to AO-metric-weighted horizontal orbital features. For individually meaningful active orbitals, these features are the differentials of normalized, inactive-projected orbital rays, weighted by $\mathbf{S}_{\mathrm{AO}}^{1/2}$. For inactive orbitals, and for the active orbitals of a complete full-AO active space, horizontal subspace tangents replace individual-ray tangents; inverse Gram factors remove dependence on the chosen basis within each subspace. Variations of the inactive projector are included in the active features.

On the nonredundant coordinates, the metric and its dual norm are

$$\mathbf{M}_o = \mathbf{U}_k^T \mathcal{L}_k^T \mathcal{L}_k \mathbf{U}_k, \qquad \|\mathbf{p}\|_{\mathbf{M}_o}^2 = \mathbf{p}^T \mathbf{M}_o \mathbf{p}, \qquad \|\mathbf{r}_o\|_{\mathbf{M}_o^{-1}}^2 = \mathbf{r}_o^T \mathbf{M}_o^{-1} \mathbf{r}_o. \tag{8}$$

We assume $\mathbf{M}_o$ is positive definite on the retained coordinates. Metric application and dual-norm evaluation need not construct a dense matrix. Local conditioning of each orbital quotient basis does not, in general, whiten the complete coupled physical metric. The metric is also distinct from a Hessian preconditioner: the former defines step size and residual norms, whereas the latter accelerates the iterative solution without changing the target equation.

2.4 Horizontal structure response and the coupled quadratic model

Collect the selected eigenvectors in $\mathbf{A}_s = (\mathbf{a}_1, \ldots, \mathbf{a}_m)$. Differentiation of $\mathbf{A}_s^T \mathbf{S} \mathbf{A}_s = \mathbf{I}$ fixes the symmetric selected-space component of their response. We choose the internal gauge

$$\dot{\mathbf{A}}_g = -\tfrac{1}{2} \mathbf{A}_s (\mathbf{A}_s^T \dot{\mathbf{S}} \mathbf{A}_s), \qquad \dot{\mathbf{a}}_s = \dot{\mathbf{a}}_{g,s} + \frac{\mathbf{Z}\mathbf{q}_s}{d_s}, \quad d_s = \sqrt{2/m}. \tag{9}$$

The columns of $\mathbf{Z}$ satisfy $\mathbf{A}_s^T \mathbf{S} \mathbf{Z} = 0$ and $\mathbf{Z}^T \mathbf{Z} = \mathbf{I}$. The latter is a coordinate convention, not orthogonality of the VB structures. Rotations within the selected equally weighted eigenspace do not affect the averaged energy and are excluded from $\mathbf{q} = (\mathbf{q}_1^T, \ldots, \mathbf{q}_m^T)^T$.

Writing $\dot{\mathbf{H}}[\mathbf{p}]$ and $\dot{\mathbf{S}}[\mathbf{p}]$ for orbital directional derivatives, differentiation of the generalized eigenproblem and projection with $\mathbf{Z}^T$ give

$$\begin{aligned} \mathbf{C}_s &= \mathbf{Z}^T (\mathbf{H} - E_s \mathbf{S}) \mathbf{Z}, \qquad \mathbf{C} = \operatorname{diag}(\mathbf{C}_1, \ldots, \mathbf{C}_m), \\ (\mathbf{B}\mathbf{p})_s &= d_s \mathbf{Z}^T \left[(\dot{\mathbf{H}}[\mathbf{p}] - E_s \dot{\mathbf{S}}[\mathbf{p}]) \mathbf{a}_s + (\mathbf{H} - E_s \mathbf{S}) \dot{\mathbf{a}}_{g,s}[\mathbf{p}] \right]. \end{aligned} \tag{10}$$

The derivative of the eigenvalue drops out because $\mathbf{Z}^T \mathbf{S} \mathbf{a}_s = 0$. The factors d_s absorb the state weights and the factor of two in the real-wave-function coefficient Hessian. The gauge term in eq (10) is essential for a moving nonorthogonal structure metric.

Let $\bar{\mathbf{A}}$ denote the orbital Hessian block after enforcing this selected-space normalization gauge and the orbital-map pullback, but before eliminating horizontal structure response. It is not the raw Hessian obtained by freezing all structure coefficients. To specify this block explicitly, define $\mathbf{K}_s = \mathbf{H} - E_s \mathbf{S}$, $\mathbf{K}_{s,u} = D\mathbf{H}[\mathbf{u}] - E_s D\mathbf{S}[\mathbf{u}]$, and $\mathbf{K}_{s,uv} = D^2\mathbf{H}[\mathbf{u}, \mathbf{v}] - E_s D^2\mathbf{S}[\mathbf{u}, \mathbf{v}]$, with E_s held at its accepted-point value in these definitions. The second variation of the normalization-constrained energy gives

$$\begin{aligned} \mathbf{u}^T \bar{\mathbf{A}} \mathbf{v} = \frac{1}{m} \sum_s \big\{ & \mathbf{a}_s^T \mathbf{K}_{s,uv} \mathbf{a}_s + 2 \dot{\mathbf{a}}_{g,s}[\mathbf{u}]^T \mathbf{K}_{s,v} \mathbf{a}_s \\ & + 2 \dot{\mathbf{a}}_{g,s}[\mathbf{v}]^T \mathbf{K}_{s,u} \mathbf{a}_s + 2 \dot{\mathbf{a}}_{g,s}[\mathbf{u}]^T \mathbf{K}_s \dot{\mathbf{a}}_{g,s}[\mathbf{v}] \big\}. \end{aligned} \tag{11}$$

All matrix derivatives in this expression include the orbital preparation map. Second derivatives of the coefficient gauge cancel against the normalization multipliers at a stationary structure solution; the first derivatives of that gauge remain. The constrained energy expansion is

$$m_k(\mathbf{p}, \mathbf{q}) = \mathcal{E}_k + \mathbf{g}_o^T \mathbf{p} + \tfrac{1}{2} \mathbf{p}^T \bar{\mathbf{A}} \mathbf{p} + \mathbf{q}^T \mathbf{B} \mathbf{p} + \tfrac{1}{2} \mathbf{q}^T \mathbf{C} \mathbf{q}, \tag{12}$$

where $\mathbf{g}_o$ is obtained from eq (3) in the coordinates of eq (5). Thus the orbital and structure blocks arise from one constrained energy, and the two coupling actions must be adjoints in their stated coordinates.

For a separated lowest-state manifold, $\mathbf{C}$ is positive definite on the horizontal space. More generally, a pseudoinverse can be used only after checking that the forcing lies in the range of $\mathbf{C}$. Eliminating structure response yields

$$\mathbf{q}(\mathbf{p}) = -\mathbf{C}^{\dagger}\mathbf{B}\mathbf{p}, \qquad \mathbf{H}_{\mathrm{rel}} = \bar{\mathbf{A}} - \mathbf{B}^{T}\mathbf{C}^{\dagger}\mathbf{B}. \tag{13}$$

Small structure gaps can make the relaxation term large even when individual coupling elements are small. A complete eigenvalue spectrum is not needed to evaluate eq (13); iterative actions on the admissible response space suffice.

2.5 Orbital restricted steps and the norm-extended formulation

We restrict the physical orbital displacement, not the joint length of orbital and structure response:

$$\min_{\mathbf{p},\mathbf{q}} m_k(\mathbf{p},\mathbf{q}) \quad \text{subject to} \quad \mathbf{p}^{T}\mathbf{M}_o\mathbf{p} \le \Delta_k^2. \tag{14}$$

The structure variables describe local electronic relaxation and do not receive an independent trust-region penalty. The stationary equations are

$$\begin{pmatrix} \bar{\mathbf{A}} + \lambda\mathbf{M}_o & \mathbf{B}^{T} \\ \mathbf{B} & \mathbf{C} \end{pmatrix} \begin{pmatrix} \mathbf{p} \\ \mathbf{q} \end{pmatrix} = - \begin{pmatrix} \mathbf{g}_o \\ \mathbf{0} \end{pmatrix}, \qquad \lambda \ge 0, \quad \lambda(\mathbf{p}^{T}\mathbf{M}_o\mathbf{p} - \Delta_k^2) = 0. \tag{15}$$

Under the assumptions of eq (13), elimination gives $(\mathbf{H}_{\mathrm{rel}} + \lambda\mathbf{M}_o)\mathbf{p} = -\mathbf{g}_o$. The exact minimum additionally requires positive semidefiniteness of the shifted relaxed Hessian. A small stationarity residual alone does not certify that curvature condition.

NEO connects a restricted Newton step to a gradient-scaled augmented eigenproblem [1]. In the eliminated orbital space, this relation reads

$$\begin{pmatrix} 0 & \alpha\mathbf{g}_o^{T} \\ \alpha\mathbf{g}_o & \mathbf{H}_{\mathrm{rel}} \end{pmatrix} \begin{pmatrix} \beta \\ \mathbf{y} \end{pmatrix} = \mu \begin{pmatrix} 1 & 0 \\ 0 & \mathbf{M}_o \end{pmatrix} \begin{pmatrix} \beta \\ \mathbf{y} \end{pmatrix}. \tag{16}$$

For $\alpha > 0$ and $\beta \neq 0$, the extracted step satisfies

$$\mathbf{p} = \frac{\mathbf{y}}{\alpha\beta}, \qquad \lambda = -\mu, \qquad (\mathbf{H}_{\mathrm{rel}} + \lambda\mathbf{M}_o)\mathbf{p} = -\mathbf{g}_o, \qquad \alpha^2 = \frac{\lambda}{-\mathbf{g}_o^{T}\mathbf{p}}. \tag{17}$$

The gradient scale is therefore tied to the restricted step rather than being an arbitrary fixed parameter. The unshifted interior Newton step is recovered as a limiting case; a zero homogeneous component or a hard case requires direct treatment of the restricted quadratic problem rather than division by β. Our solver obtains the small projected restricted step spectrally and uses eqs (16) and (17) to express its norm-extended interpretation. It does not repeatedly solve a full-dimensional augmented eigenproblem or form $\mathbf{H}_{\mathrm{rel}}$.

2.6 Independent response spaces and full residual certification

Orbital and structure spaces are expanded independently. With basis matrices $\mathbf{V}$ and $\mathbf{W}$, the trial response is $\mathbf{p} = \mathbf{V}\mathbf{u}$, $\mathbf{q} = \mathbf{W}\mathbf{v}$. The projected blocks are

$$\begin{aligned} &\mathbf{a} = \mathbf{V}^T\bar{\mathbf{A}}\mathbf{V}, \qquad \mathbf{b} = \mathbf{W}^T\mathbf{B}\mathbf{V}, \quad \mathbf{c} = \mathbf{W}^T\mathbf{C}\mathbf{W}, \quad \mathbf{m} = \mathbf{V}^T\mathbf{M}_o\mathbf{V}, \\ &\mathbf{h} = \mathbf{a} - \mathbf{b}^T\mathbf{c}^\dagger\mathbf{b}. \end{aligned} \tag{18}$$

New directions require new operator images; previously computed images are retained at the same accepted orbital point. Changing the trust radius does not invalidate these images. The projected Schur complement is inexpensive once these blocks are available, but is not assumed to reproduce the full structure relaxation merely because its projected equations are solved accurately.

Certification uses residuals in the original retained coordinate spaces:

$$\begin{aligned} \mathbf{r}_o &= \mathbf{g}_o + \bar{\mathbf{A}}\mathbf{p} + \mathbf{B}^T\mathbf{q} + \lambda\mathbf{M}_o\mathbf{p}, \\ \mathbf{r}_s &= \mathbf{B}\mathbf{p} + \mathbf{C}\mathbf{q}, \\ \left(\|\mathbf{r}_o\|^2_{\mathbf{M}_o^{-1}} + \|\mathbf{r}_s\|_2^2\right)^{1/2} &\le \epsilon_{\text{abs}} + \eta_k\|\mathbf{g}_o\|_{\mathbf{M}_o^{-1}}. \end{aligned} \tag{19}$$

The Euclidean structure norm refers specifically to the scaled coordinates in eq (9). Step feasibility and complementarity are checked separately. Curvature information from the projected space is not a general certificate of positive semidefiniteness outside that space.

A small structure residual can still give a significant orbital error when $\mathbf{C}$ is ill-conditioned. Where needed, we solve for a response correction and examine its orbital effect:

$$\mathbf{C}\delta\mathbf{q} = -\mathbf{r}_s, \qquad \mathbf{r}_o^{\text{corr}} = \mathbf{r}_o + \mathbf{B}^T\delta\mathbf{q}. \tag{20}$$

An unresolved correction supplies an additional response direction; a sufficiently accurate correction completes the candidate response. This targets the response needed by the current orbital candidate instead of solving every orbital basis direction to final response accuracy. It does not eliminate coupling work: appending an orbital direction still requires its orbital-to-structure image.

2.7 Trial evaluation, approximate models, and local convergence

Only the orbital component of the local step is applied in eq (5). At the trial orbitals, the selected structure problem is solved again; $\mathbf{q}$ is an auxiliary local response, not a replacement for this nonlinear reoptimization. Acceptance compares the actual relaxed energy decrease with the coupled-model prediction,

$$\rho_k = \frac{\mathcal{E}_k - \mathcal{E}(\mathbf{x}_k + \mathbf{U}_k\mathbf{p})}{m_k(\mathbf{0},\mathbf{0}) - m_k(\mathbf{p},\mathbf{q})}. \tag{21}$$

Accepted points receive a new true gradient, chart, and local model. Rejected trials retain the accepted-point operator data while the trust radius is adjusted. No cross-iteration secant Hessian history is required by this NEO formulation; a previous orbital direction may provide a warm start.

The full coupled model must be distinguished from the production core-only approximation. The latter retains direct inactive/core response and the nonlinear pullback with fixed upstream adjoints, but omits both local active response and horizontal structure response. It is therefore not

simply eq (12) with $\mathbf{B} = 0$, nor is it the complete fixed-structure orbital Hessian. The current admission decision compares response-space dimension and estimated contraction work with orbital-space dimension and core work. This is a computational work heuristic, not an error bound on omitted curvature. In both branches, trial energies and accepted gradients are evaluated for the same relaxed VBSCF objective. Full and core-only calculations must be identified separately in the numerical results.

Near a smooth isolated minimum with a nonsingular quotient Hessian, the full model approaches an ordinary Newton step once the trust region is inactive. Superlinear local convergence requires the effective Newton residual to be $o(\|\mathbf{g}_o\|)$; quadratic convergence requires an $O(\|\mathbf{g}_o\|^2)$ residual, locally Lipschitz second derivatives, and correspondingly controlled structure-solver and derivative errors. A forcing sequence $\eta_k = O(\|\mathbf{g}_o\|)$ supplies the relative part of this requirement, but a fixed nonzero ϵ_{abs} eventually limits the certified accuracy. These local conclusions do not automatically apply to the core-only model or establish rapid entry into the Newton neighborhood.

One-electron orbital blocks are used as a preconditioner, not substituted for the full Hessian in these statements. Block-LBFGS and TNHVP are comparison methods: the former retains secant history with a block initial inverse, whereas the current TNHVP implementation uses a block-LBFGS baseline with conditional HVP-based correction. Their performance must be measured against NEO rather than inferred from the labels "first order" and "second order."

3 IMPLEMENTATION

3.1 Matrix-free derivative actions

Describe the accepted-point data required for gradient and HVP evaluation: normalization, inactive projection, integral transformation, structure action, and response. Identify reused quantities and directional quantities, without claiming that matrix-free means no large intermediates. Give memory costs in orbital, structure, and unique-string dimensions.

3.2 Trial steps and accepted-point updates

Give the acceptance and trust-radius update settings associated with eq (21). State which quantities are refreshed at an accepted point and which are reused during subspace expansion. Count rejected trials as work even when they do not advance the accepted-iteration counter.

Algorithm 1. Accepted-point second-order VBSCF optimization (outline)

1. Read the same orbital guess and structure list used by all comparison methods.
2. Solve the selected structure problem; evaluate energy and the physical orbital gradient.
3. Build the support-preserving local coordinates, metric, derivative actions, and preconditioner.
4. Expand the iterative second-order model; certify its residual and record response treatment.
5. Retract the orbital step, solve the trial structure problem, and evaluate model agreement.
6. Accept or reject the trial; refresh accepted-point data and test both stopping criteria.

3.3 Cost accounting

Report orbital HVPs, orbital-to-structure and transpose-coupling actions, structure H/S actions, trial evaluations, and accepted steps separately. Include setup, final observables, and total process memory in end-to-end measurements; isolate optimizer time in an additional breakdown.

4 COMPUTATIONAL DETAILS

4.1 Methods and matched conditions

Method	Selector	Role
XMVB-style L-BFGS	ISCF=5	Conventional quasi-Newton baseline
TNHVP	ISCF=7	Matrix-free truncated Newton comparison
Block-LBFGS	ISCF=8	Enhanced baseline with one-electron block initial inverse
NEO	ISCF=9	Principal norm-extended second-order method

Table 1. Comparison methods. Input selectors are implementation details, not method names.

Use identical initial orbitals, geometry, basis, active electrons/orbitals, spin, selected structure list, state weights, integral representation, and final convergence tests. Compare exact and RI only in separate matched groups. Dense and Davidson structure solvers must solve the same selected generalized eigenproblem to controlled accuracy.

Record the projected-gradient convention and the physical dual-metric norm. The working energy-change and projected-gradient targets are 10^{-7} hartree and 10^{-3}, respectively; confirm the precise implemented norms and final settings before collection. Use an independently tighter calculation for endpoint comparisons.

4.2 Test-set design

Group	Examples	Purpose
Small derivative oracles	F_2; small selected-structure cases	Finite differences, dense Hessian, endpoint checks
Localized HAO	241 (benzene), 240, 7975, 7963, YAMSAI	Support constraints and convergence variation
Transition metals	MnF_2 and $FeCl_2$	Conditioning, spin, and OEO/HAO behavior
Larger response spaces	CERRAS and LOFLEA	Response cost and memory

Group	Examples	Purpose
Equal-weight state average	Small multistate cases; selected pyrrole/OEO cases	Root-space and response stability

Table 2. Proposed test groups; molecular specifications and eligibility remain to be frozen.

Do not include unresolved endpoint discrepancies as successful validation. A PySCF CASSCF comparison is a variational-space reference only when the OEO structure expansion, spin/symmetry sector, integrals, and core convention match; otherwise it is a diagnostic comparison.

4.3 Hardware and reproducibility

Collect publication timings on Hanhai25 compute nodes through Slurm. Report CPU model, allocated cores, affinity, OpenMP/BLAS threads, compiler flags, dependency versions, and commit identifier. Record peak RSS at the job/process level. Preserve nonconverged runs and resource failures in the dataset rather than dropping them.

5 RESULTS AND DISCUSSION

5.1 Derivative and response validation

Table 3 will report gradient directional differences, HVP directional differences, bilinear symmetry defects, and coupled-versus-Schur agreement on small cases. Include nonorthogonal well-conditioned and near-singular orbital points, HAO and OEO, and at least one equal-weight state average. [Insert verified numerical errors.]

5.2 Local convergence

Figure 2 will show energy error and physical gradient norm for identical controlled perturbations around a common stationary point. Establish the local convergence regime separately from the number of iterations required to enter it. [Insert trajectories and the residual forcing sequence.]

5.3 End-to-end comparison

Figures 3 and 4 will plot accepted steps and total wall time against a common system ordering, with failures visibly marked. Table 4 will provide final energies, energy changes, physical gradients, action counts, wall time, and RSS. Discuss cases where fewer steps do not yield lower wall time. [Insert matched results; do not mix revisions.]

5.4 Where the second-order cost is spent

Figure 5 will break down setup, core HVP, coupling/structure response, trial evaluation, and final output. Analyze larger active spaces using measured work and memory rather than accepted-step count alone. Compare full and restricted response with the model treatment stated for every run. [Insert stage timings and response statistics.]

5.5 Ablations and limitations

Separate model changes from preconditioner changes. Prioritize matched full/restricted response and one-electron block/Hessian-diagonal comparisons; identity preconditioning need not be a production benchmark. Discuss small structure gaps, incomplete selected-VB spaces, root changes, and redundant-coordinate conditioning. [Insert only ablations that answer a specific algorithmic question.]

6 CONCLUSIONS

Summarize the validated second-order formulation, the measured range of efficiency, and the cost of structure relaxation. State which method is recommended for which workload only if the data support that distinction. [Write the quantitative conclusion after Sections 5.1–5.5 are complete.]

ASSOCIATED CONTENT

Supporting Information

Planned contents: coordinate and response derivations; update-map derivatives; finite-difference protocols; complete molecular specifications and initial orbitals; raw convergence trajectories; non-convergence records; timing scripts and environment details. Submit the SI separately from the main manuscript.

AUTHOR INFORMATION

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DATA AVAILABILITY STATEMENT

[Supply a persistent public repository link or DOI for the input files, initial orbitals, raw outputs, benchmark metadata, and analysis scripts. Identify the code release and commit used for every figure. Explain any justified restrictions.]

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[Complete the VBSCF/XMVB, inexact-Newton, trust-region, L-BFGS, and density-fitting literature before submission. This starter list is not an exhaustive priority review.]